

Johannes Nguyen

IT SPECIALIST FOR SYSTEM INTEGRATION · DKFZ HEIDELBERG

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Profile

IT Specialist for System Integration (German IHK qualification, 2026) at the DKFZ, focused on Linux, automation and Kubernetes. I pick up new technologies quickly and take them from the first comparison to production: evaluate the options, present them to the team, implement, hand over. I trace faults systematically to their root cause instead of just treating symptoms, a way of working I learned as an automotive mechatronics technician.

Experience

German Cancer Research Center (DKFZ)

Heidelberg, DE

IT SPECIALIST FOR SYSTEM INTEGRATION (ODCF)

2022 – present

- **Responsibility:** took workload off senior colleagues and increasingly owned projects, regularly asked for **technical recommendations**
- **Cluster support (ODCF):** support and troubleshooting for the W610 HPC cluster running **IBM LSF**, user management, shares and quotas on **IBM ESS**
- **From evaluation to production:** compared alternatives (e. g. AWS vs. Ascender), presented them to the team, implemented and handed over
- **Delivered projects:** **AWX** and **Ascender** on **Kubernetes** (k3s), **Bitwarden** on-prem, **Harbor** registry, **EasyBuild** module system with GCC 14, **Grafana Alloy** and per-user CPU/RAM limits
- **Identity and access:** connected Harbor and AWX to **LDAP/AD**; implemented Bitwarden SSO via **ADFS** (SAML) since Entra ID with SCIM was ruled out by licensing, tested **Pocket ID** as an OIDC alternative
- **Automation:** own **Ansible** roles and playbooks, integrated new services into the existing deploy automation, host deployment via Kickstart
- **Logging:** built central logging with **Loki** and Grafana, rolled out the **Grafana Alloy** log agent cluster-wide, analysed and fixed proxy and permission issues during the rollout
- **Documentation:** wrote more than 10 **SOPs** for recurring procedures, including ESS shares and quotas, host deployment, TLS certificates and LSF maintenance
- **Own RAG system:** made SOPs and notes searchable for LLM assistants through hybrid search behind an **MCP server**, to shorten onboarding and switch between tasks more reliably
- **Incident handling:** during a staff shortage took over two **RStudio** servers with several hundred users, resolved the licensing outage without prior experience and kept them running
- **Datacenter:** independent on-site work, prepared **NetBox** as the source of truth for cabling on behalf of the network team (own trace and report tools, 11 racks), moved servers to fibre, **Proxmox** network planning with Ceph and LACP

Vögele

Ludwigshafen, DE

COMMERCIAL VEHICLE MECHANIC

2021 – 2022

BASF, Porsche, TRW, GRN

Various locations

INTERNSHIPS: COMPUTER SCIENCE, PROTOTYPING, MATERIALS TESTING

2013 – 2016

Education

German Cancer Research Center (DKFZ)

Heidelberg, DE

IT SPECIALIST FOR SYSTEM INTEGRATION (IHK APPRENTICESHIP), COMPLETED 07/2026

2022 – 2026

- **Final project:** set up a **Proxmox Backup Server** and integrated it into the ODCF Proxmox environment, including the VxRail nodes without local storage
- Backup concept, LDAP integration, hardening with firewalld, monitoring via Prometheus exporter, LACP uplink and operations documentation for the team
- **Rotations:** Service Center (automated client installation with WDS), IT department (Proxmox, Matrix42, emergency environment during a security incident), then ODCF

Mercedes-Benz

Mannheim, DE

AUTOMOTIVE MECHATRONICS TECHNICIAN (APPRENTICESHIP)

2017 – 2021

Projects

Homelab

Personal project

PROXMOX VE · PANGOLIN · TAILSCALE · POCKET ID · PBS

- Three-node Proxmox cluster with around 15 services (GitLab, Nextcloud, Immich, Vaultwarden, Harbor and others), GPU passthrough and backups via **Proxmox Backup Server**
- Access through Cloudflare and a **Pangolin** gateway with CrowdSec and per-node tunnel failover, SSH only via **Tailscale**, SSO via **OIDC** with Pocket ID

git-iac: Self-hosted GitOps

github.com/j551n-ncloud

GITLAB CI · ANSIBLE · HARBOR · TRIVY

- Separate build and deploy pipelines: images tagged with the commit SHA in Harbor and scanned with **Trivy**, a merge request bumps the pin, the merge rolls out via Ansible after a Proxmox snapshot

systemd-resource-control

Work project ODCF

ANSIBLE · SYSTEMD · PAM

- Ansible role that limits CPU and RAM per user on shared cluster nodes through systemd slices, applied at login via PAM, in production use

Skills

System administration	Linux (RHEL, Debian, systemd, auditd), Windows Server, Bash, IBM LSF, Dell iDRAC
Automation	Ansible (own roles and playbooks), AWX, GitLab CI/CD, GitHub Actions, Kickstart
Containers	Kubernetes (k3s, Helm, operators), Docker Compose, Harbor
Virtualization	Proxmox VE and Backup Server (HA, GPU passthrough, SDN), VMware ESXi
Storage & networking	IBM ESS, Ceph, NFS, VLANs, LACP, NetBox, Tailscale
Identity & security	LDAP/AD, ADFS, OIDC/SSO, CrowdSec, Trivy, TLS/PKI
Monitoring & logging	Loki, Grafana Alloy, Grafana, Checkmk, Prometheus
Development & AI	Python (FastAPI), SQL, TypeScript, Node.js, RAG, MCP, LLMs